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RESEARCH INTERESTS

Multi-agent coordination mechanisms, uncertainty quantification, and reliability in high-stakes AI systems. Investigating how task decomposition affects reasoning quality and how agents coordinate effectively in complex domains. PhD-bound Fall 2026.

EDUCATION

Northeastern University, Toronto, ON

Expected May 2026

Master of Science in Analytics | GPA: 3.96/4.0 | Concentration: Applied Machine Intelligence
Relevant Coursework: Machine Learning, Deep Learning, Statistical Methods, Data Mining

University of Nottingham, UK

December 2015

Master of Science in Entrepreneurship, Innovation & Management
Thesis: "Management of Open Innovation Process in the Food Sector"
Research Focus: Coordination mechanisms across organizational boundaries, innovation systems

Sir Sandford Fleming College, Toronto, ON

July 2024

Graduate Certificate in Project Management | GPA: 3.88/4.0
Focus: Project planning, risk management, stakeholder coordination, delivery execution

University for Development Studies, Tamale, Ghana

December 2013

Bachelor of Arts in Economics & Entrepreneurship Studies

RESEARCH EXPERIENCE & FOUNDATION PROJECTS

Self-Healing Data Pipeline Platform | Multi-Agent Causal Reasoning Research

AI-native infrastructure investigating coordination mechanisms and causal inference
January 2026 | GitHub: github.com/CrillyPienaah/self-healing-pipelines

- Research Platform: Built autonomous system that detects and remediates data pipeline failures using GPT-4 multi-agent architecture (Detective → Fixer → Critic), achieving 89.8% average confidence across 21 test scenarios with 7.7s generation time
- Three Research Contributions:
 - Causal Reasoning Dataset: 21+ labeled (failure, context, fix) triples for investigating whether LLMs distinguish root causes (upstream API changes) from correlations (schema drift)
 - Hierarchical Multi-Agent Framework: Detective → Fixer → Critic architecture enabling research on RL-optimized coordination vs. rule-based orchestration
 - Zero-Shot Code Generation: Production-ready fix generation with confidence scoring, rollback planning, and quality evaluation framework
- Key Research Questions:
 - Can LLMs perform causal inference in software systems beyond pattern matching?
 - Can hierarchical RL policies optimize agent coordination better than hand-designed rules?
 - How do we formalize code quality evaluation for autonomous systems?

- Technical: FastAPI + LangChain + GPT-4 + React dashboard, 100% test coverage, <50ms API latency, comprehensive monitoring and audit trails for EU AI Act compliance
- Performance: 40% reduction in manual maintenance time, <10min resolution vs. 2-4hr manual, production-ready code quality validated through systematic testing
- Research Positioning: Foundation for 3 potential NeurIPS/ICML papers on causal reasoning, hierarchical RL, and zero-shot code generation. Dataset expansion to 1,000+ examples planned Q2 2026
- Next Steps: Implementing RL-based agent orchestration, academic paper submissions, OpenAI Residency application with production dataset and empirical findings

LuminaMed-AI v2.0 | Multi-Agent Clinical Intelligence Platform

Research Foundation: Investigating coordination mechanisms in multi-agent reasoning systems

Live Deployment: [Radiology Portal](#) | [Patient Portal](#)

- Designed 6-agent architecture (triage, imaging analysis, differential diagnosis, treatment planning, safety validation, coordination manager) to test hypothesis that task decomposition improves reliability in medical AI reasoning
- Key Research Findings:
 - Explicit agent communication improved diagnostic accuracy from 87% → 94%
 - Task decomposition reduced edge-case hallucination by 40%
 - Identified systematic failure patterns in agent handoffs and coordination breakdowns
 - Discovered emergent coordination patterns not explicitly designed into the system
- Open Research Questions Generated:
 - Why do certain decomposition strategies generalize better than others?
 - What is the minimal coordination structure needed for reliable multi-agent systems?
 - Can we predict which medical cases will trigger multi-agent failure modes?
 - Are performance gains from better information flow or simply additional compute?
- Technical Implementation: LangGraph multi-agent orchestration, Google Gemini 2.5 Flash, FastAPI backend, dual-portal frontend (Streamlit + Next.js), PostgreSQL, FHIR R4 compatibility, clinical-grade safety features with explicit uncertainty quantification
- Impact Metrics: 92% diagnostic accuracy, 8% hallucination rate (down from ~30% without verification), 16-20s report generation, audit trails for regulatory compliance
- Next Steps: Technical report in progress documenting coordination mechanisms and failure mode analysis. Foundation for PhD research direction.

SME Growth Co-Pilot | Agent Specialization Study

Experimental investigation of specialization vs. generalization tradeoffs in multi-agent systems

Google Kaggle AI Agents Competition: Top 50 of 11,494 participants

- Research Question: Do specialized agents outperform general-purpose agents on complex business analysis tasks requiring multi-domain expertise?
- Experimental Design: Built 6-agent system with specialized roles (financial analysis, market research, operational efficiency, customer insights, competitive intelligence, strategic synthesis) to test performance across different business analysis scenarios
- Key Findings:
 - Specialization improved domain-specific accuracy but reduced adaptability

- Agent coordination overhead increased with specialization
- Optimal agent granularity appears task-dependent, not universal
- Surfaced fundamental questions about task boundary definition in multi-agent systems
- Technical: Multi-agent orchestration, real-time data integration from multiple sources, performance monitoring, A/B testing framework for coordination protocols
- Performance: 100% success rate on test cases, 1ms average response time, identified \$135K revenue opportunities through gap analysis that manual consulting missed
- Research Insight: This project revealed that the "right" level of agent specialization depends on task complexity, domain boundaries, and coordination costs—not a one-size-fits-all answer

Daavi | Trust Mechanisms in Low-Resource Markets

Exploration: Can AI fundamentally change trust dynamics in markets with weak institutions?

- Research Context: African diaspora face 68% trust barrier in real estate investment due to fraud (30-40% of transactions) and information asymmetry. Built platform to investigate whether transparency mechanisms and AI verification can establish trust where institutional trust is low.
- Approach: Implemented multi-layer verification (government registry checks, legal document validation, physical inspection workflows) combined with fraud detection algorithms and transparency features (verification history, agent track records)
- Key Insight: Trust in AI systems isn't purely technical—it requires understanding institutional context, cultural factors, and human decision-making under uncertainty. Technical solutions must account for societal constraints.
- Research Questions Emerged:
 - How do AI recommendation systems need to adapt for different institutional contexts?
 - What's the relationship between transparency mechanisms and user trust in AI?
 - Can algorithmic verification substitute for institutional trust, or only complement it?
- Technical: Next.js, NestJS, PostgreSQL/PostGIS for spatial analysis, recommendation algorithms balancing relevance with fraud risk, anti-fraud pattern detection (95% accuracy)
- Impact: 85% onboarding completion rate, addresses major barrier in \$5B+ diaspora investment market
- Built from Canada while coordinating verification in Ghana—forced robust, transparent systems design that works across distance

RESEARCH BACKGROUND

MSc Thesis: Management of Open Innovation Process in the Food Sector

University of Nottingham, UK | 2015

- Investigated how food sector companies manage open innovation—balancing internal R&D with external knowledge sources, managing IP boundaries, coordinating across organizational lines
- Methodology: Mixed-methods research combining qualitative interviews with industry practitioners and quantitative analysis of innovation outcomes
- Key Contribution: Analyzed how organizational structure, incentive systems, and coordination mechanisms affect innovation success. Documented systematic patterns in coordination failures and success factors.
- Research Skills Developed: Systematic literature review, empirical study design, qualitative and quantitative data analysis, academic writing, systems thinking about coordination mechanisms
- Connection to Current Research: My innovation management research asked "How do organizations coordinate across boundaries to innovate?" My AI research asks "How do agents coordinate across task boundaries to reason?" Same fundamental question about coordination mechanisms, different domain.

Transition to AI Research

- Applied research methodology from Nottingham thesis to AI systems: systematic experimentation, ablation studies, failure mode analysis, hypothesis testing, rigorous documentation
- Built production systems as research platforms to generate empirical questions about intelligence, not just to ship products
- Combined project management discipline (Fleming) with research methodology (Nottingham) and technical implementation (Northeastern) to execute complex research-oriented projects

MACHINE LEARNING PROJECTS (ACADEMIC PORTFOLIO)

Customer Churn Prediction | Banking Analytics

- Built Random Forest classifier achieving 86.93% accuracy on 10,000 customer records
- Discovered regional variance: 45% churn rate in Germany vs. 16% overall—recommended region-specific retention strategies
- Applied K-Means clustering for customer segmentation (Low/Medium/High risk groups)
- Focus: Feature engineering, model selection, actionable business insights from ML

Bank Marketing Propensity Model | Predictive Analytics

- Developed XGBoost model (AUC 0.93) predicting term deposit conversions on 11,162 records
- Created automated pipeline demonstrating 3x improvement in conversion through propensity-based targeting
- Feature analysis revealed: call duration and campaign history >> demographics as predictors
- Focus: Probability calibration, ROC analysis, business impact quantification

PROFESSIONAL EXPERIENCE

ICON Leadership Institute, Toronto, ON

May 2025 – Dec 2025

AI/ML Product Strategist & Innovation Lead

- Built custom CRM system from scratch, automating lead tracking and reducing manual work 80%
- Optimized GPT-4 prompts for document summarization: 20% relevance improvement, 15% cost reduction through systematic prompt engineering and evaluation
- Automated processing of 450K daily records via Python + SQL pipelines (10 hours → 1 hour)
- Created Tableau dashboards translating raw data into strategic insights for executive decision-making
- Implemented data quality validation ensuring 99.9% accuracy across multiple data sources

Chrisline Prints & Advertising, Kumasi, Ghana

May 2014 – June 2023

Founder & Technical Lead

- Built Python + SQL data pipelines centralizing data for 50+ SME clients (30% improvement in report accuracy)
- Created Excel VBA dashboards increasing campaign ROI 30% through better targeting and timing optimization
- Led 10+ process improvement workshops achieving team adoption of automation systems
- Managed full business operations: sales, operations, technical development, client relationships

TECHNICAL SKILLS

AI/ML Research & Development:

- Multi-Agent Systems: LangChain, LangGraph, agent orchestration, coordination protocols
- LLMs & APIs: GPT-4, Claude API, Gemini 2.5 Flash, prompt engineering, fine-tuning

- ML Fundamentals: scikit-learn, XGBoost, Random Forest, Logistic Regression, K-Means, PCA
- Advanced Techniques: RAG systems, vector databases, uncertainty quantification, ablation studies

Full-Stack Development:

- Backend: FastAPI, NestJS, Node.js, REST APIs, WebSockets
- Frontend: Next.js, React, responsive design
- Auth & Security: JWT, RBAC, encryption standards
- Integrations: Stripe payments, Slack API, webhook systems

Data Engineering & Analysis:

- Databases: PostgreSQL, MySQL, SQL Server, PostGIS (spatial)
- Data Processing: Python (pandas, NumPy), ETL pipelines, data warehousing, star schemas
- Visualization: Tableau, Power BI, custom dashboards
- Cloud: AWS, Azure, Vercel deployment, Docker containers

Research & Project Management:

- Research Methods: Experimental design, hypothesis testing, systematic analysis, mixed-methods
- Project Execution: Risk management, stakeholder coordination, agile delivery, documentation
- Version Control: Git/GitHub, collaborative development, CI/CD pipelines

CERTIFICATIONS & RECOGNITION

- Top 50 of 11,494 participants, Google Kaggle AI Agents Competition (December 2025)
- 5-Day AI Agents Intensive, Kaggle + Google (December 2025)
- AWS Generative AI Certification, Udacity (2025)
- Applied AI & No-Code Development, DTTP (2025)
- Google Data Analytics Professional Certificate (2024)

PUBLICATIONS & RESEARCH WORK

- "Agent Coordination Mechanisms in Clinical Reasoning: LuminaMed-AI Case Study" — Technical report in progress, 2025
- "Management of Open Innovation Process in the Food Sector" — MSc Thesis, University of Nottingham, 2015

RESEARCH TRAJECTORY & AVAILABILITY

PhD-bound Fall 2026, pursuing doctoral research in multi-agent AI systems with focus on coordination mechanisms, reliability, and reasoning in high-stakes domains.

Applying for OpenAI Residency 2026 to gain mentorship from frontier researchers before doctoral work. Six months of focused research experience would significantly accelerate trajectory toward becoming an independent AI researcher.

Research Approach: Build production systems to understand where intelligence breaks, apply rigorous methodology to investigate fundamental questions, combine technical implementation with conceptual thinking about mechanisms and systems.

Available June 2026 for research residencies, PhD programs, or research-oriented AI/ML roles.